3. System Requirements

3.1. System Features

3.1.1. Calendar View

3.1.1.1. Introduction/Purpose of Feature

This feature allows Schedulers to view all of their tasks planned for any given month or day. When an individual day is selected, each of the Scheduler's tasks for the specified day is displayed including their start and end times.

3.1.1.2. Stimulus/Response

Stimulus: The Scheduler navigates to the Calendar View page.

Response: The Calendar View page displays the current year and current month in a monthly calendar format.

Stimulus: The Scheduler specifies a particular date in the Calendar View.

Response: The system retrieves all tasks for the specified date from the database with their correct start and end times listed. The system also allows the Scheduler to perform Task Management for a task of their choice.

Stimulus: The Scheduler uses the "Previous Month" option.

Response: The Calendar View page displays the previous month's Calendar View in a monthly calendar format.

Stimulus: The Scheduler uses the "Next Month" option.

Response: The Calendar View page displays the next month's Calendar View in a monthly calendar format.

3.1.1.3. Associated Functional Requirements

3.1.1.3.1. Monthly Calendar View *(O: Bragg, M_1: Allen)*

The application shall display the following:

- A specified calendar month's days in a monthly Gregorian calendar format
- A visible marker on all days which have at least one task assigned to them
- An option to view the preceding month’s Calendar View
- An option to view the succeeding month’s Calendar View

3.1.1.3.2. Daily Calendar View *(O: Bragg)*

The application shall display the specified calendar day's tasks sorted by their start times.

3.1.1.3.3. Daily Calendar Task Interaction *(O: Bragg, M_1: Allen)*

The application shall allow the Scheduler to choose from the following Task Management operations for a specific calendar day:

- Task editing
- Task addition
- Task removal

3.1.1.3.4. Previous Month Option *(O: Allen)*

The application shall provide the capability to display the Monthly Calendar View for the preceding month's days.

3.1.1.3.5. Next Month Option *(O: Allen)*

The application shall provide the capability to display the Monthly Calendar View for the succeeding month's days.

3.1.1.3.6. Previous Month Handling for January *(O: Allen)*

The application shall display the preceding year's December's Monthly Calendar View if a December Monthly Calendar View is displayed.

3.1.1.3.7. Next Month Handling for December *(O: Allen)*

The application shall display the succeeding year's January's Monthly Calendar View if a January Monthly Calendar View is displayed.

3.1.2. Task Management

3.1.2.1. Introduction/Purpose

This feature allows Schedulers to edit, add, and remove tasks from their schedule.

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3.1.2.2. Stimulus/Response

Stimulus: The Scheduler uses the "Add Task" option on a Daily Calendar View page.

Response: A Task Management form for creating tasks will be displayed.

Stimulus: The Scheduler uses the "Edit Task" option on a Daily Calendar View page.

Response: A Task Management form for editing tasks will be displayed.

Stimulus: The Scheduler uses the "Delete Task" option on a Daily Calendar View page.

Response: The Scheduler is prompted if they are sure they wish to remove the task. When confirmation is provided, the task is removed from the Calendar View and the Gold Star database.

3.1.2.3. Associated Functional Requirements

3.1.2.3.1. "Add Task" Task Form *(O: Bragg, M_1: Allen)*

The system shall provide the capability to specify the following properties for a new task:

- Task name
- Optional task description of up to 300 ASCII characters
- Task priority
- Task start time
- Task end time
- Task nonnegotiable status

3.1.2.3.2. "Edit Task" Task Form *(O: Bragg, M_1: Allen)*

The system shall provide the capability to change the following properties for an existing task specified by the Scheduler:

- Task name
- Optional task description of up to 300 ASCII characters
- Task priority
- Task start time
- Task end time
- Task nonnegotiable status

3.1.2.3.3. Task Form Validation *(O: Bragg, M_1: Allen)*

The system shall validate each inputted task property for the following criteria before a task is added or edited:

- The task name contains valid text.
- The task description contains valid text or no text.
- The task priority matches a valid predefined priority level.
- The task start time is a valid time.
- The task end time is a valid time.
- The task start time comes before the task end time.
- The Scheduler specified a valid task nonnegotiable status.

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3.1.2.3.4. Remove Task *(O: Bragg, M_1: Allen)*

The system shall ensure a specified task is removed from the following after the Scheduler's approval:

- The Scheduler's Calendar View
- The Gold Star database

3.1.3. Task Prioritization

3.1.3.1. Introduction/Purpose of Feature

While creating or editing tasks, Task Prioritization enables Schedulers to categorize their scheduled tasks based on prioritized importance levels. Task Prioritization allows individual Schedulers' calendars to display tasks in order of criticality.

3.1.3.2. Stimulus/Response Sequences

Stimulus: The Scheduler specifies a task to manage.

Response: The system prompts the Scheduler to specify priority levels for the given task using a Task Management form.

Stimulus: The Scheduler specifies a priority level during task configuration.

Response: The current priority level choice is displayed on the Task Management form.

Stimulus: The Scheduler finishes Task Management.

Response: The task and its corresponding priority level are validated. The task and priority level are then saved to the Gold Star database.

3.1.3.3. Associated Functional Requirements

3.1.3.3.1. Task Priority Levels *(O: Allen)*

The system shall provide the capability to classify tasks in the following Priority Levels:

- Low
- Medium
- High

3.1.4. Daily Mood Check-In

3.1.4.1. Introduction/Purpose

Daily Mood Check-In allows Schedulers to record their current mood when using Gold Star. This feature helps Schedulers reflect on their emotional state while managing daily tasks. Schedulers can select a mood which best represents how they are feeling.

3.1.4.2. Stimulus/Response Sequence

Stimulus: The Scheduler selects a mood for daily check-in.

Response: Gold Star saves the selected mood for the Scheduler's daily check-in.

3.1.4.3. Associated Functional Requirements

3.1.4.3.1. Daily Mood Check-In *(O: Nebuchadnezzar)*

The system shall allow the Scheduler to specify a mood for their daily check-in.

3.1.4.3.2. Store Daily Mood Check-In *(O: Nebuchadnezzar)*

The system shall save the mood specified by the Scheduler for their daily check-in.

3.1.4.3.3. Edit Daily Mood Check-In *(O: Nebuchadnezzar)*

The system shall allow the Scheduler to edit their mood throughout the day.

3.1.5. Adaptive Scheduling

3.1.5.1. Introduction/Purpose

Adaptive Scheduling allows Schedulers to adjust their daily schedules when unexpected events occur, or when they are unable to complete planned tasks. Unfinished tasks can be rescheduled to another date, allowing Schedulers to maintain flexibility without penalization for missed tasks. The system also adjusts proposed daily schedules based on the Scheduler's selected productivity level, the Scheduler’s mood, and task priorities to manage daily responsibilities.

3.1.5.2. Stimulus/Response Sequences

Stimulus: The Scheduler selects an unfinished task to reschedule.

Response: The system allows the Scheduler to move the unfinished task to another day.

Stimulus: The Scheduler selects a new date for the unfinished task.

Response: The system updates the task's scheduled date to reflect the Scheduler's selection.

Stimulus: The Scheduler submits their daily Productivity level.

Response: The system adjusts the proposed daily schedule based on the Scheduler's selected productivity level.

Stimulus: The Scheduler reviews the adjusted daily schedule.

Response: The system displays the proposed schedule for the Scheduler to review.

3.1.5.3. Associated Functional Requirements

3.1.5.3.1. Unfinished Task Rescheduling (*O: Nebuchadnezzar, M_1: Looney, M_2: Allen*)

The system shall allow the Scheduler to reschedule an unfinished task to a different date.

3.1.5.3.2. Productivity-Based Schedule Adjustment (*O: Nebuchadnezzar, M_1: Looney*)

The system shall adjust the proposed daily task schedule based on the Scheduler's selected productivity level.

3.1.6. Habit Stacking

3.1.6.1. Introduction/Purpose of Feature

Habit Stacking helps Schedulers reduce decision fatigue by associating new tasks with existing routines. By connecting new tasks and established tasks, Schedulers can organize their larger schedule into smaller isolated sequences of tasks called Habit Stacks. Habit Stacks help the Scheduler visually identify which task should be completed next in a given sequence. This feature aims to reduce task overload and encourage consistency when completing daily routines.

3.1.6.2. Stimulus/Response Sequences

Stimulus: The Scheduler selects an existing task to associate with another task.

Response: The system allows the Scheduler to associate both tasks into a small Habit Stack sequence of tasks.

Stimulus: The Scheduler saves the Habit Stack.

Response: The system stores new changes to a Habit Stack into the Gold Star database.

Stimulus: The Scheduler views a task that belongs to a specified Habit Stack.

Response: The system displays the specified task from the specified Habit Stack.

Stimulus: The Scheduler removes a task's association with a specified Habit Stack.

Response: The system removes the specified task's association with the Habit Stack through Gold Star's database.

3.1.6.3. Associated Functional Requirements *(O: Looney)*

3.1.6.3.1. Habit Stack Creation *(O: Looney)*

The system shall allow the Scheduler to create an isolated sequence (i.e., Habit Stack) of currently saved tasks.

3.1.6.3.2. Saving Habit Stack *(O: Looney)*

The system shall store any changes made to the Habit Stack's sequence of tasks.

3.1.6.3.3. Display Habit Stack *(O: Looney)*

The system shall display all associated tasks from a Habit Stack in the Scheduler's sequential order.

3.1.6.3.4. Habit Stack Modification *(O: Looney)*

The system shall allow the Scheduler to change the task sequence within an existing Habit Stack.

3.1.6.3.5. Habit Stack Removal *(O: Looney, M_1: Allen)*

The system shall allow the Scheduler to remove one or more specified tasks within a specified Habit Stack.

3.1.6.3.5. Habit Stack Deletion *(O: Allen)*

The system shall allow the Scheduler to remove one of their saved Habit Stacks.

3.2. Performance Requirements

3.2.1. Responsiveness

3.2.1.1. Webpage Load Time *(O: Brackett, M_1: Looney)*

The Gold Star homepage shall finish loading within 3 seconds of the Scheduler requesting the page.

3.2.1.2. Calendar Loading *(O: Brackett, M_1: Looney)*

The system shall display the Calendar View within 3 seconds of the Scheduler navigating to the calendar page.

3.2.1.3. Calendar Updates *(O: Brackett, M_1: Looney)*

The system shall save and display calendar changes within 5 seconds of the Scheduler submitting an update.

3.2.1.4. Account Creation *(O: Brackett)*

The process for creating an account should have an average completion time of 4 seconds from the time the Scheduler's details are submitted.

3.2.1.5. Account Login *(O: Brackett)*

The account login process should be completed within 3 seconds of verifying the Scheduler's credentials.

3.2.2. Throughput

3.2.2.1. Scheduler Session Handling *(O: Brackett)*

The webpage should be capable of supporting up to a minimum of 450 Scheduler sessions simultaneously with no significant loss in performance.

3.2.2.2. Load Handling *(O: Brackett)*

The webpage server should maintain a throughput of 1,000 requests per second without dropping connections or rejecting requests.

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3.2.3. Execution Time

3.2.3.1. Database Execution *(O: Allen)*

All database queries shall be executed within 40 milliseconds to account for task validation time, network latency, and network processing overhead.

3.2.4. Storage Capacity

3.2.4.1. Database Storage Capacity *(O: Allen)*

The database shall have a minimum storage capacity of 20 GB to fit Scheduler data and task data.

3.3. Design Constraints

3.3.1. Compatibility Requirement

3.3.1.1. Browser Compatibility *(O: Allen)*

The system shall be accessible via a web Graphical User Interface on the following modern web browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Brave
- Safari

3.3.2. Technical Constraints

3.3.2.1. Database Simulation *(O: Allen)*

The system shall simulate remote database queries using an active local MySQL relational database management system.

3.3.2.2. Server Simulation *(O: Allen)*

The system shall simulate remote web server activity using an active local Node.js runtime environment listening on port 3000.

3.3.2.3. Scheduler Account Simulation *(O: Allen)*

The system shall simulate three predefined Scheduler accounts using an active local MySQL relational database management system for testing Scheduler-related performance metrics.

3.4. Software System Attributes

3.4.1. Reliability

3.4.1.1. Mean Time Between Failures *(O: Allen)*

The system should maintain a minimum mean time between failures (MTBF) of 8,000 hours to avoid excessive service downtime.

3.4.2. Availability

3.4.2.1. Uptime Rate *(O: Allen)*

The system should maintain a minimum monthly availability rate of 99.9% to ensure consistent service.

3.4.3. Security

3.4.3.1. SQL Injection Mitigation *(O: Allen)*

The system shall use prepared statements on all database queries to avoid SQL Injection attacks.

3.4.4. Maintainability

3.4.4.1. Coding Convention Guidelines *(O: Allen)*

The codebase shall follow language-specific standard guidelines to ensure uniform formatting and effortless maintenance.

3.4.4.2. Documentation Guidelines *(O: Allen)*

All code documentation must:

- Be well-documented for ease of maintenance
- Be well-preserved with virtual backups to account for hardware failures

3.4.5. Portability

3.4.5.1. Operating System Run Compatibility *(O: Allen)*

The system should run on the following operating systems:

- Windows
- Linux
- OS/X
- Android
- iOS

3.4.5.2. Operating System Build Compatibility *(O: Allen)*

The system should build on the following operating systems:

- Windows
- Linux
- OS/X
- Android
- iOS

3.5. Other Requirements

3.5.1. Accessibility *(O: Allen)*

The system shall comply with the Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to accommodate Schedulers' accessibility needs.

3.5.2. Data Privacy *(O: Looney)*

The system shall restrict access to a Scheduler's stored task information to that Scheduler's account.

3.5.3. Copyright Compliance *(O: Looney, M_1: Allen)*

All third-party images, icons, and other assets included in Gold Star's development shall be used in accordance with their applicable licenses.